ADC Initialization Overview

Overview & Implementation Steps

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- ADC1 Configuration Function
- ADC Analog Watchdog Initialization Function
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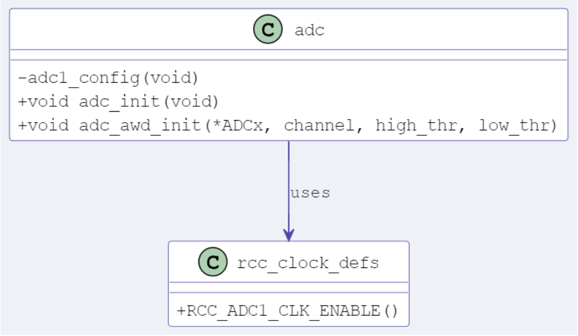
ADC Initialization Overview

Overview: The ADC implementation contains an initialization function for the required ADC configuration(s) and configuring the Analog Watchdog for setting high/low thresholds.

- **ADC Init**: Enables the ADC clock, enables the temperature sensor, enables ADC interrupts and calls the ADC1 configuration function.



- **ADC1 Configuration**: Enables EOC interrupt, configures timer triggering, sets sampling time and specifies the temperature sensor channel, and enables the ADC.

- **ADC AWD Init**: Sets high/low thresholds, configures the temperature sensor channel, enables the AWD and AWD interrupts.

- Project Significance: Provides a simple interface for initializing ADC configurations, e.g., obtaining internal MCU temperature, and configuring the Analog Watchdog (AWD).

ADC Initialization Function

ADC Initialization Function Steps

- ✓ **ADC1 Clock Enable**: Using the RCC_ADC1_CLK_ENABLE() macro, enable the ADC1 clock connected to the APB2 bus.
- ✓ **Set ADC Prescaler**: Configure the ADC clock by setting the prescaler to 4 in the CCR, resulting in PCLK2 of 45 MHz / 4 = 11.25 MHz.
- ✓ **Enable Temperature Sensor**: Enable the temperature sensor in the CCR.
- ✓ **ADC1 Configuration**: Call the ADC1 configuration function (specific settings, timer triggers, etc., to be covered next).
- ✓ **Enable ADC Interrupts**: Enable the ADC interrupt in the NVIC.
- ✓ Project Significance: The adc_init() function provides a comprehensive ADC initialization and configuration by setting up common settings and calling the adc1_config() function.

ADC Configuration Function

ADC Configuration Function Steps

- ✓ **Clear any Previous Configurations**: Set CR1 and CR2 to zero.
- ✓ **Enable EOC Interrupt**: Set the EOC (End Of Conversion) interrupt in CR1.
- ✓ **Configure Trigger**: Setup trigger detection on rising edge and Timer 2 as external trigger in CR2.
- ✓ **Setup EOCS**: Configure EOC flag in CR2 to be set at the end of each regular conversion.
- ✓ **Set Sampling Time**: Configure sampling time for temp sensor to 112 cycles in SMPR1.
- ✓ **Set Conversion Sequence**: Setup Channel 18 (Temperature Sensor) in SQR3.
- ✓ **Enable ADC**: Enable the ADC by setting the ADON bit in CR2.

- ✓ Project Significance: The adc1_config() function sets up ADC1 for internal temperature sensor measurements with specific settings for the timer trigger, EOC, sampling time, and conversion sequence, ensuring accurate and timely temperature readings.

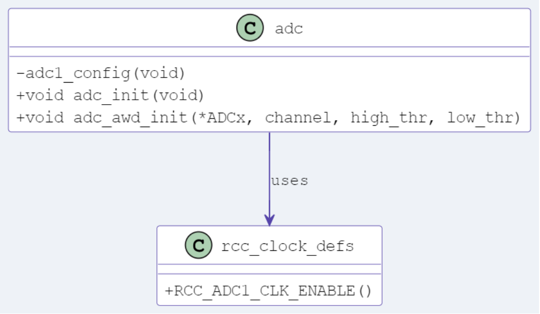
ADC Analog Watchdog Function

ADC AWD Initialization Function Steps

- ✓ **Set Thresholds**: Configure high and low thresholds in HTR and LTR.
- ✓ **Select Channel**: Set watchdog for Channel 18 (Temperature Sensor) in CR1.
- ✓ **Enable AWD**: Enable AWD for single channel in CR1.
- ✓ **Enable AWD Interrupt**: Set AWD interrupt in CR1.
- ✓ Project Significance: The adc_awd_init() function configures the ADC's Analog Watchdog to monitor the temperature sensor channel with specified high and low thresholds, ensuring timely interrupts for out-of-range values.

ADC Implementation Steps

Overview: The ADC implementation includes functions for configuring the ADC to read temperature values from the MCU's internal sensor and setting up the Analog Watchdog for out-of-range triggers.



- Objective: Prepare adc.c and adc.h files to support temperature readings and Analog Watchdog functionality.

- Key Steps:
  - Create adc.h: Define ADC channels and function prototypes.
  - Create adc.c: Implement function definitions.

- Project Significance: Establishes a simple interface for ADC initialization, meeting the System Health Monitor's requirements.

Next: ADC Header File